

Emir K Ulusoy Portfolio



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Summary

- Patent holder for the LTE PIM Catcher — a big data and machine learning method for detecting passive intermodulation from network data instead of a site visit. US Patent #10009784 ([View Patent](#)).
- Rebuilt capacity breach thresholds on dynamic logic drawn from cell footprint, user behavior and measured customer impact — accessibility and throughput — instead of fixed counters. False alarms went away, prioritization sharpened, and upgrade spend now lands only where customers feel it. The method guides **millions in annual network investment**.
- Built a Linux-based 5G, LTE and Wi-Fi test tool on iPerf and FTP, covering indoor point-to-point and outdoor GPS testing, in place of a commercial tool at **\$15K per license**. Metrics come over SSH from a Cradlepoint R1900, throughput from curl and iperf across TCP and UDP, and Wi-Fi state is parsed from nmcli and iwconfig.
- Moved indoor and outdoor coverage reporting onto QGIS and PyQGIS using the qchainage and mmqgis modules, cutting **\$5K per user** in license cost.
- Redesigned the 5G geolocation pipeline around degree-based grid aggregation, matching stored precision to what GPS can actually resolve. Storage fell 100x, and the analysis window went from three weeks to two years — over **500M samples**.
- Wrote a Python converter that turns Planet coverage prediction TIFFs into polygons, so geo-filtering and analysis run free in QGIS. Removed a paid license dependency worth **\$2K per user**.
- Built Python and JavaScript automation for real-time diagnostics, letting field teams triage before rolling a truck. Fewer revisits and manlift rentals, over **\$100K** a year across **20+ field teams**.

Big Data and Machine Learning Patent

Patents Assigned to YUPANA, INC.

Remote detection and analysis of passive intermodulation problems in radio base stations

Patent number: 10009784

Abstract: Systems and methods are described for configuring a base station node. A set of base station parameters are received by a remote control unit. The remote control unit receives and installs dynamic parameter values for base station setup selected by a server application based on the received base station parameters. The remote control unit also receives dynamic parameter values for network integration generated by the server application via the network connection, the dynamic parameter values for network integration being generated in response to a user selection. The remote control unit may then configure the base station node using the dynamic parameter values for network integration. When configuration is complete, the remote control unit may transmit an indication to the server application, and finalizes integration of the base station node in response to a request from the server application.

Type: Grant

Filed: June 1, 2017

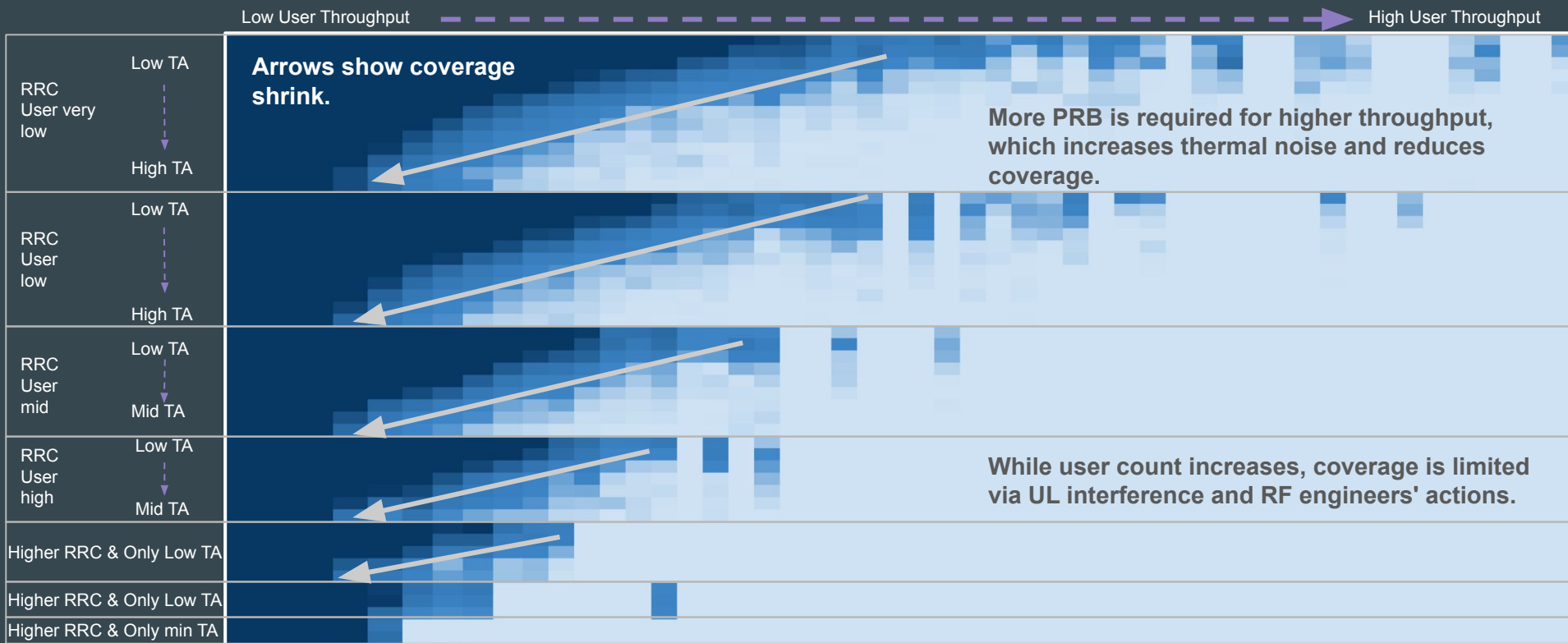
Date of Patent: June 26, 2018

Assignee: YUPANA, INC.

Inventors: Hakan Evircan, Emir Kursad Ulusoy, Muzaffer Mete Dalan

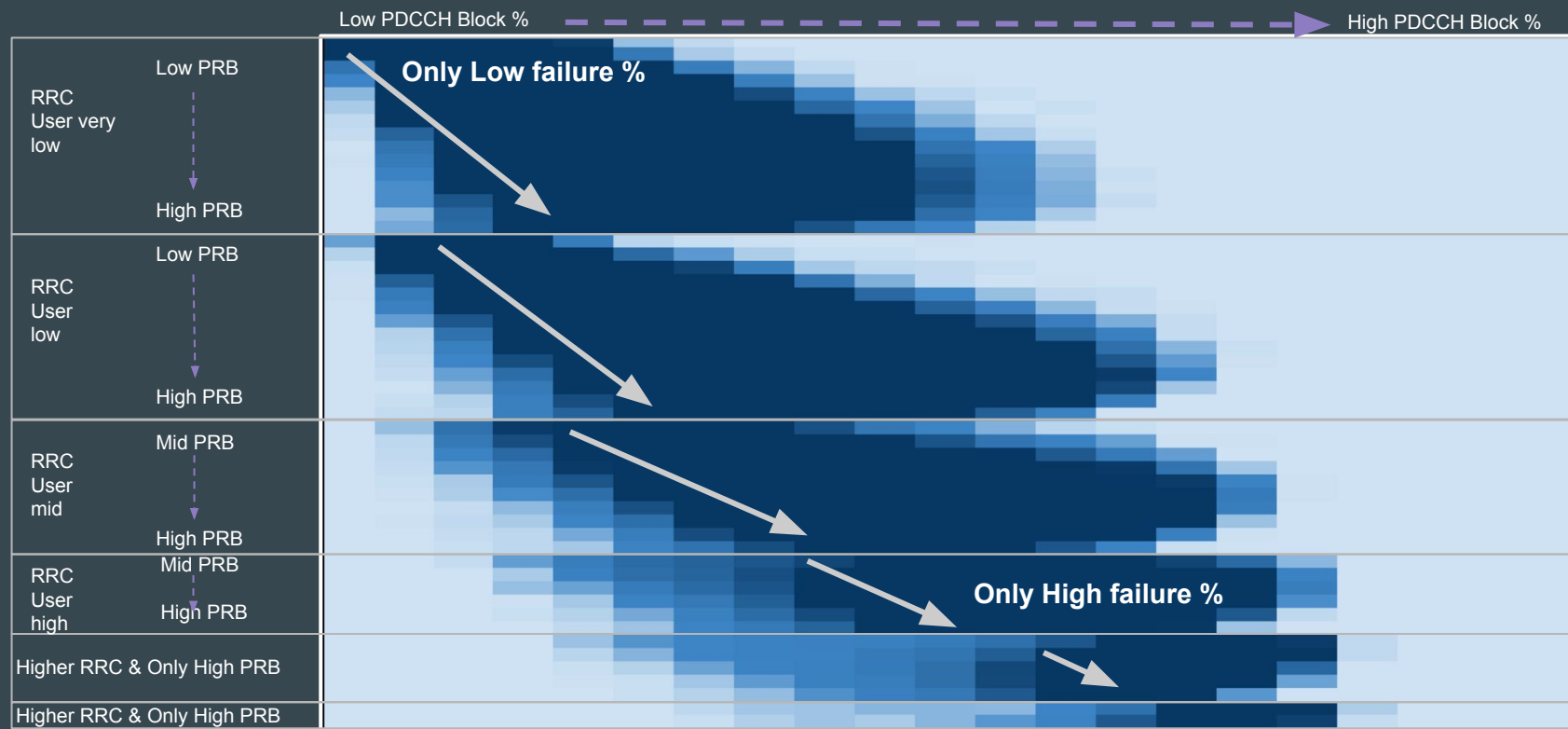
Demand side: where coverage actually ends – [link](#)

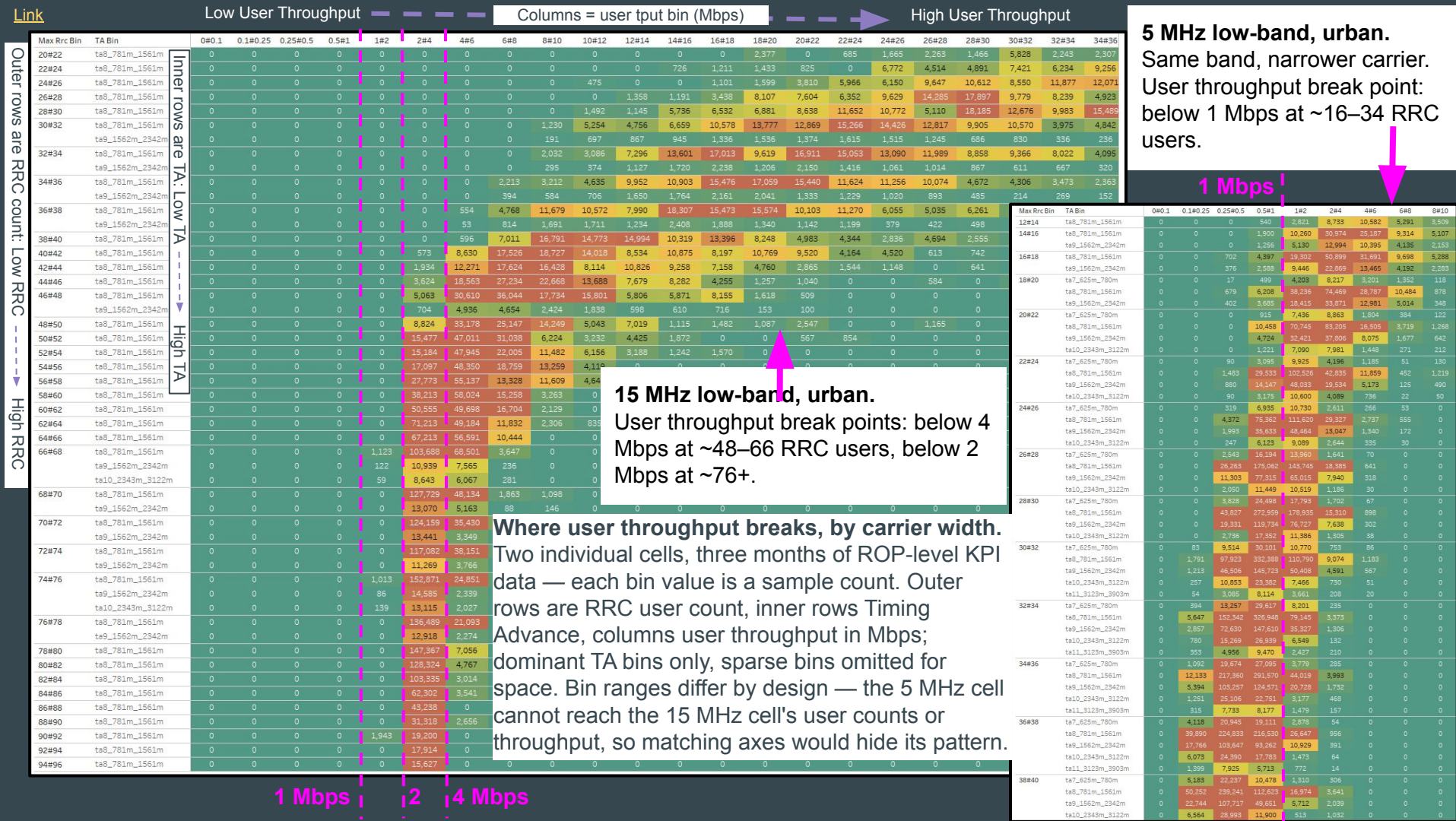
Cross-referencing RRC users, timing advance and user throughput. Higher throughput needs more PRB, which raises thermal noise and pulls coverage in. The link budget says where the cell edge should be; this shows where it is.



Supply side: the scheduler runs out before the spectrum – [link](#)

RRC users against PRB utilization and PDCCH blocking. Blocking climbs with user count while PRB still has headroom, so the scheduler sets the ceiling before the spectrum does. It shows up earliest on narrow bandwidth cells.





Linux tool for analyzing WiFi performance on end-user devices I

Developed an open-source, Linux-based Wi-Fi testing and reporting toolkit using Python, bash scripting, and QGIS. No expensive licenses required.

Signal Level/Quality

Utilizing the "nmcli" to scan every WiFi channel that is available and gather all information:

```
nmcli dev wifi rescan  
sudo nmcli -t --fields  
NAME,SSID,BSSID,MODE,<...>,  
ACTIVE dev wifi
```

Collecting TPUT

FTP version: Based on the performance, curl has been selected after evaluating wget, axel, and aria2c options:

```
curl -u <user>:<pass> -o  
<filename> <ftp directory file>
```

*UDP/TCP version uses iPerf commands:

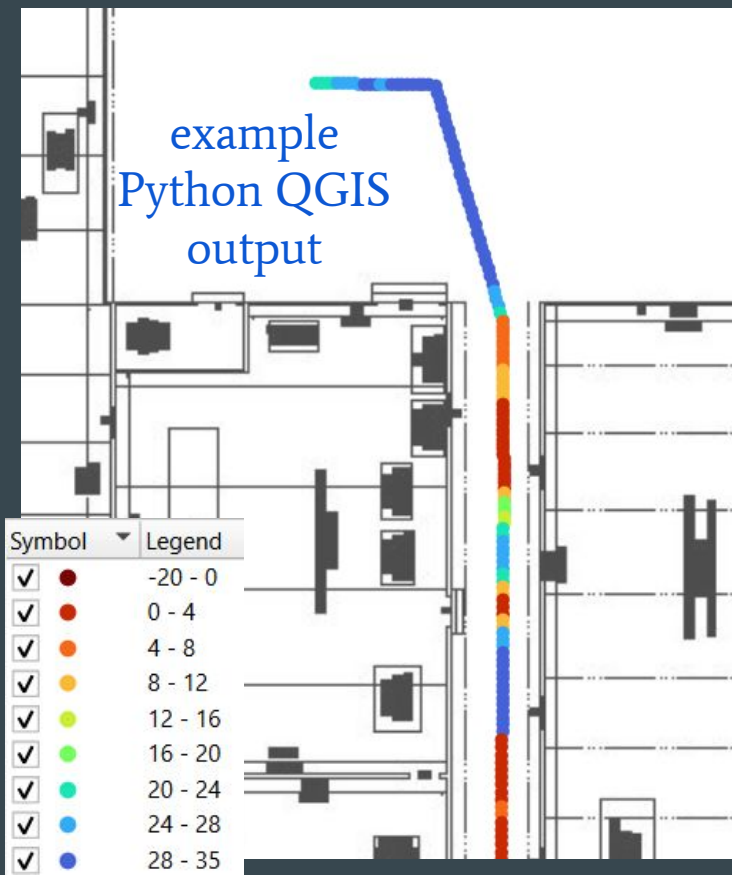
```
iperf3 -c <server> -t <time>
```

Python Parse/Merge

Using Python to combine and parse data and prepare a CSV file to create maps.

See next slide.

Linux tool for analyzing WiFi performance on end-user devices II



Example indoor test Map Output

Built for teams running Wi-Fi surveys regularly. Python and bash for collection, QGIS for the heat map, no commercial license anywhere in the chain.

Where it stands:

About \$5K per engineer against the commercial equivalent

Report generation scripted end to end

Same output format on every project

Used on more than ten indoor deployments

GitHub links:

[Wi-Fi Data Collection](#)

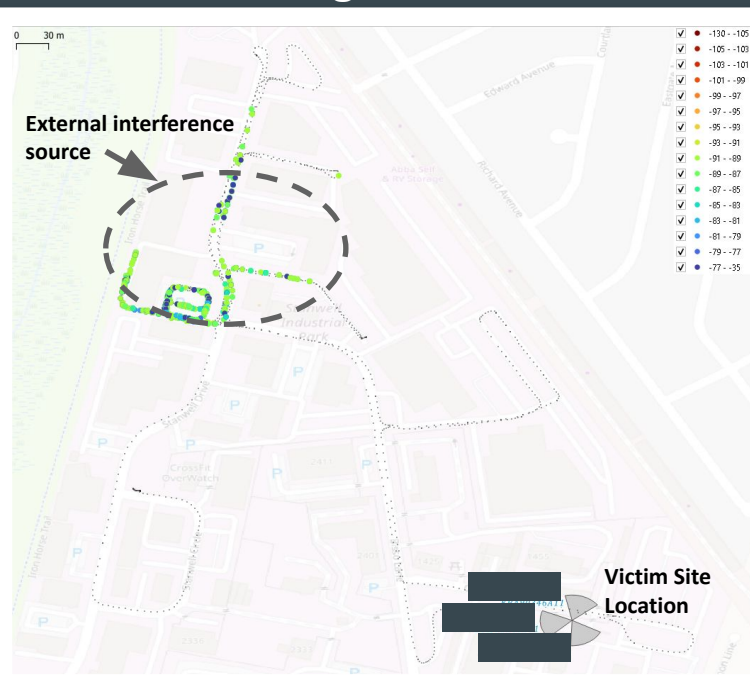
[Indoor Heat Map Reporting](#)

[Linkedin post](#)

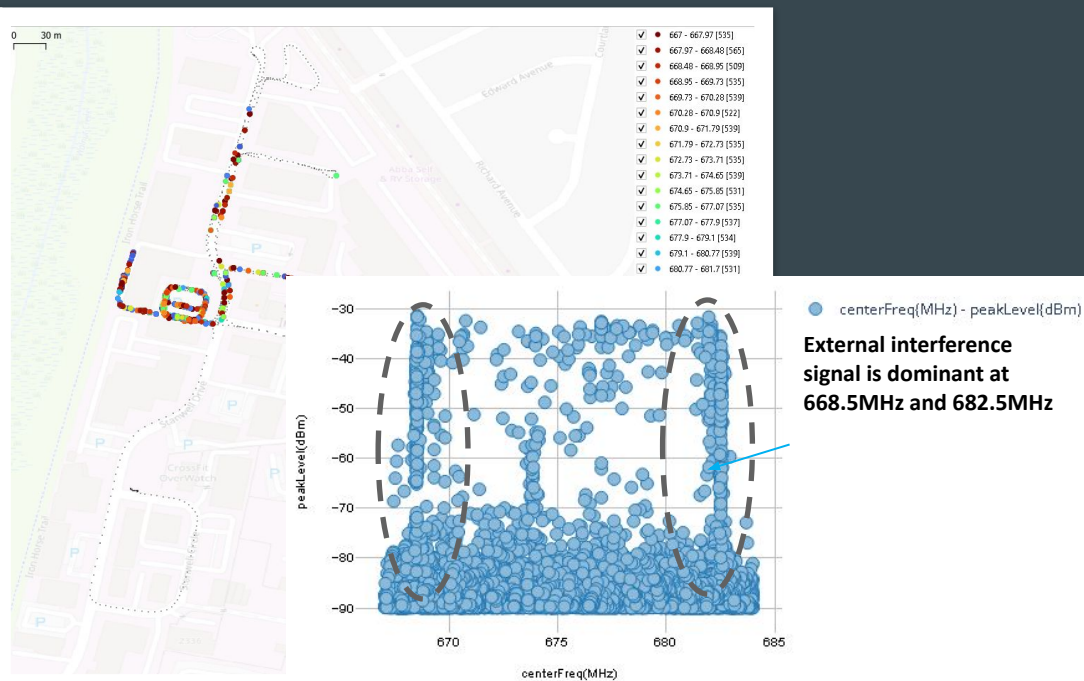
External Interference Testing – [link](#)

A mapping-capable spectrum analyzer with an integrated display and GPS runs into the tens of thousands. This setup came in under \$6K and produced the frequency against level plot below, which placed the interferer at 668.5 and 682.5 MHz.

External Inter. Signal Level



External Inter. Frequency



Linux tool for 5G/LTE performance via Cradlepoint R1900 Modem

```
[R1900 /]$ get status/wan/devices.  
TE\|CUR_PLMN\|EMMCOMMSTATE\|EMMSTATE\|EMMSUB:  
"ACTIVE_SCELL0": "Active",  
"BANDWIDTH_SCELL0": "20 MHz",  
"BAND_SCELL0": ,  
"CELL_ID": ,  
"CHANNEL_SCELL0": ,  
"CUR_PLMN": ,  
"DLFRQ": ,  
"EMMCOMMSTATE": "RRC Connected",  
"EMMSTATE": "Registered",  
"EMMSUBSTATE": "Normal Service",  
"FRQ_SCELL0": ,  
"LTEBANDWIDTH": "20 MHz",  
"MODEMPSSTATE": "Attached",  
"MODEMSYSMODE": "LTE",  
"PHY_CELL_ID": ,  
"RFBAND": ,  
"RFCHANNEL": ,  
"RSRP": "-45",  
"RSRQ": "-10",  
"SINR": "30.0",  
"SRVC_TYPE": "LTE",  
"SRVC_TYPE_DETAILS": "None",  
"TXCHANNEL": ,  
"TX_LTE": "-32.0",  
"ULFRQ": ,
```

Example 5G/LTE SSH connection

Through an SSH connection, 5G/LTE data (serving cell, quality, signal level, band, etc.) is gathered over the Cradlepoint R1900 Modem.

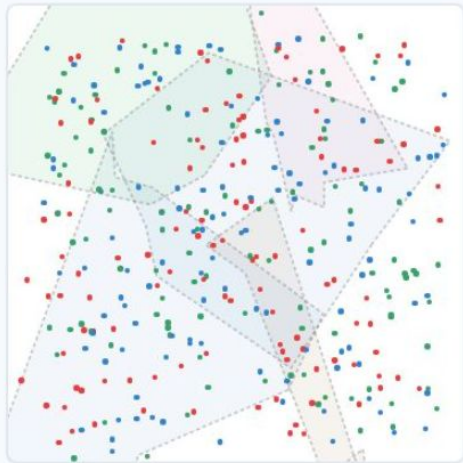
```
ssh -v -oHostKeyAlgorithms=+ssh-rsa <user>@<device id>
```

```
get status/wan/devices/.../diagnostics | grep  
CELL_ID\|PHY_CELL_ID\|RSRP\|RSRQ\|SINR\|...
```

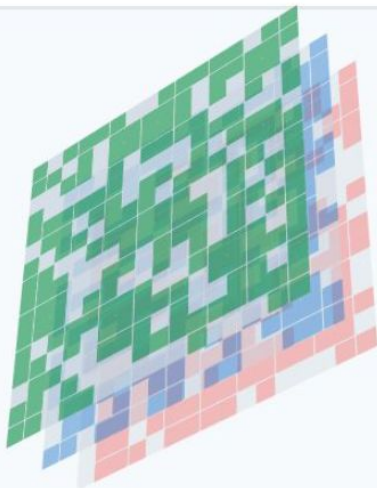
Storing precision that does not exist – [link](#)

5G geolocation was stored at six decimal places, about 0.1 m, against a GPS accuracy of 5 to 10 m. Four of those decimals were noise, paid for in storage that grew with every month of data kept. Re-aggregated on a degree-based grid in AWS Redshift at four decimals, roughly 11 m, with CRS transformations removed and linear KPI averaging applied to RSRP, SINR and RSRQ.

✗ BEFORE



✓ AFTER



Result

Storage: 180 GB to 2 GB

Analysis window: 3 weeks to 2 years, about 500M samples

Polygon geometry out of the processing path

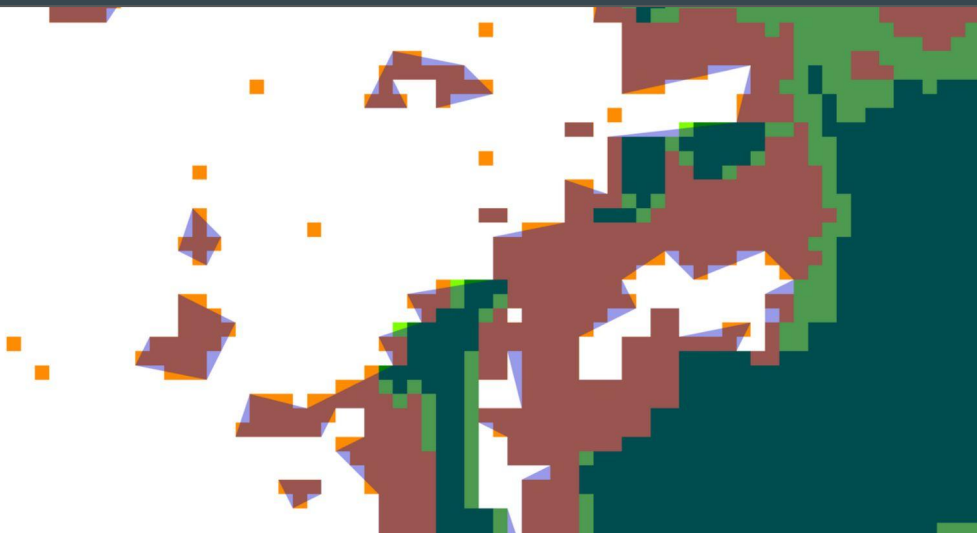
More samples per bin, so more statistical confidence

Planet Coverage TIFF file to GeoPackage converter – [link](#)

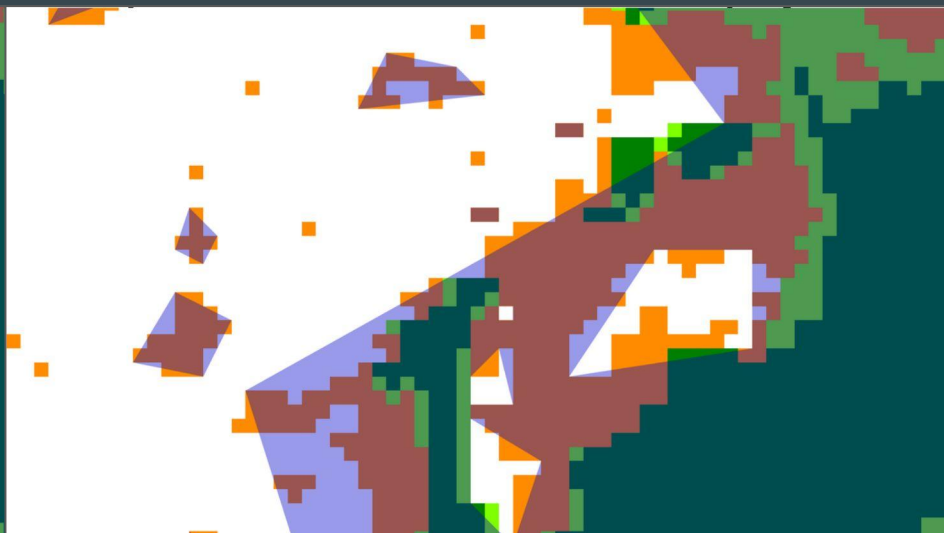
Open-source TIFF to GeoPackage converter for Planet prediction output. Removed about \$2K per user in annual GIS licensing, put several departments on the same prediction workflow, and runs roughly five times faster at national scale.

GitHub [link](#)

10m resolution for mission-critical processes



100+ m resolution for zoom-out view



Green and orange polygons are prediction tool output. The black, half-transparent polygon is arranged as a resolution output.

Web-based reporting tool over Google Apps Script platform – [link](#)

Javascript // Chart // Auto report

Apps Script Example

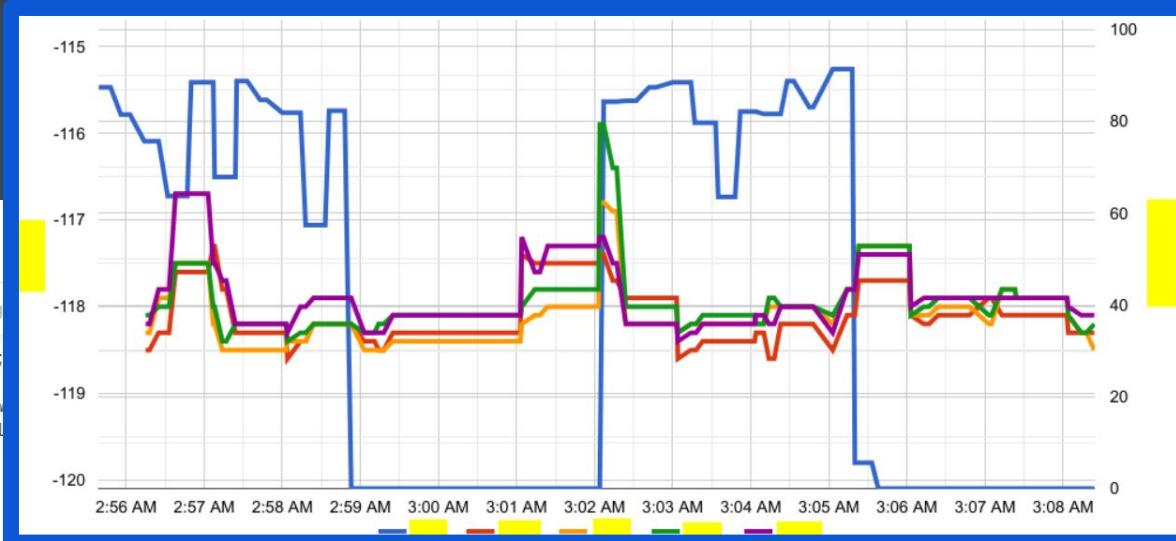
Files

- Code.gs

Libraries

Services

```
435 for(var k =0;
436 {
437   data.addRow
438     [r]+'_'+col
439 }
440 data.build();
441
442 var chartBuilder = Charts.newLinechart() //chart options
443 .setDataTable(data)
444 .setOption('width', 1000)
445 .setOption('height', 500)
446 .setOption('chartArea.left', 75)
447 .setOption('chartArea.width',850)
448 .setOption('chartArea.height',400)
449 .setOption('legend.position', 'bottom')
450 .setOption('legend.maxLines', 5)
451 .setOption('legend.textStyle.fontSize', 12)
452 .setOption('legend.textStyle.color', 'black')
453 .setOption('title', ".... -- " +celllist1[r] + " -- " + testday1 )
454 .setOption('titlePosition', 'out')
455 .setOption('titleTextStyle.italic',1)
```



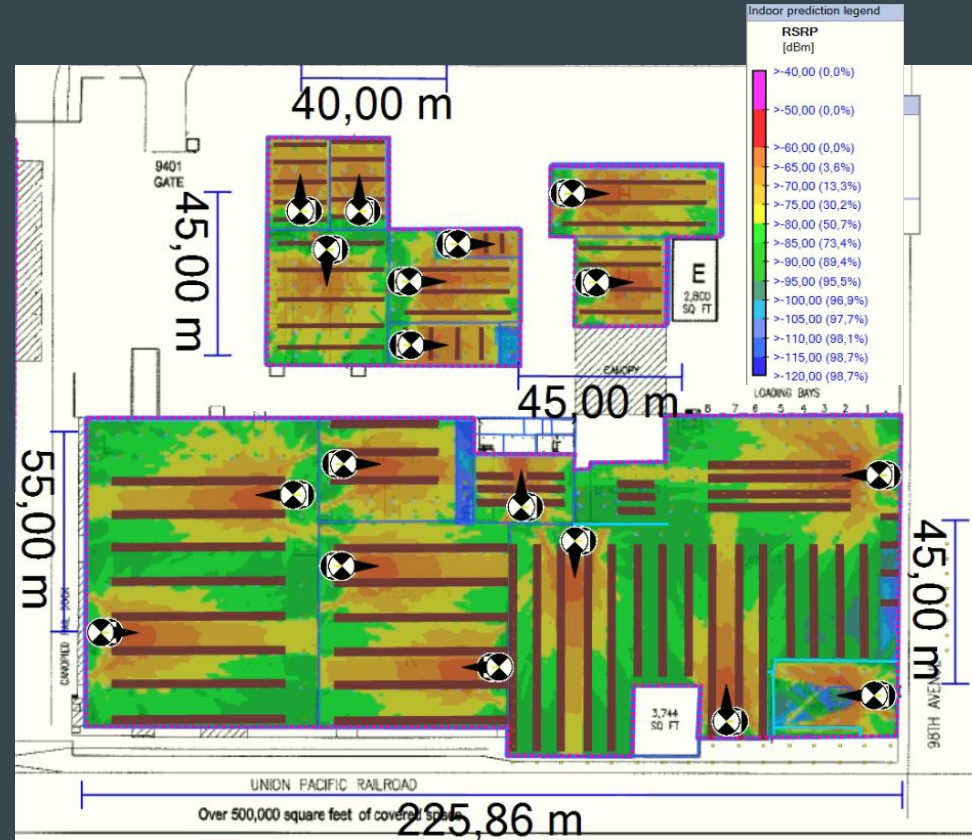
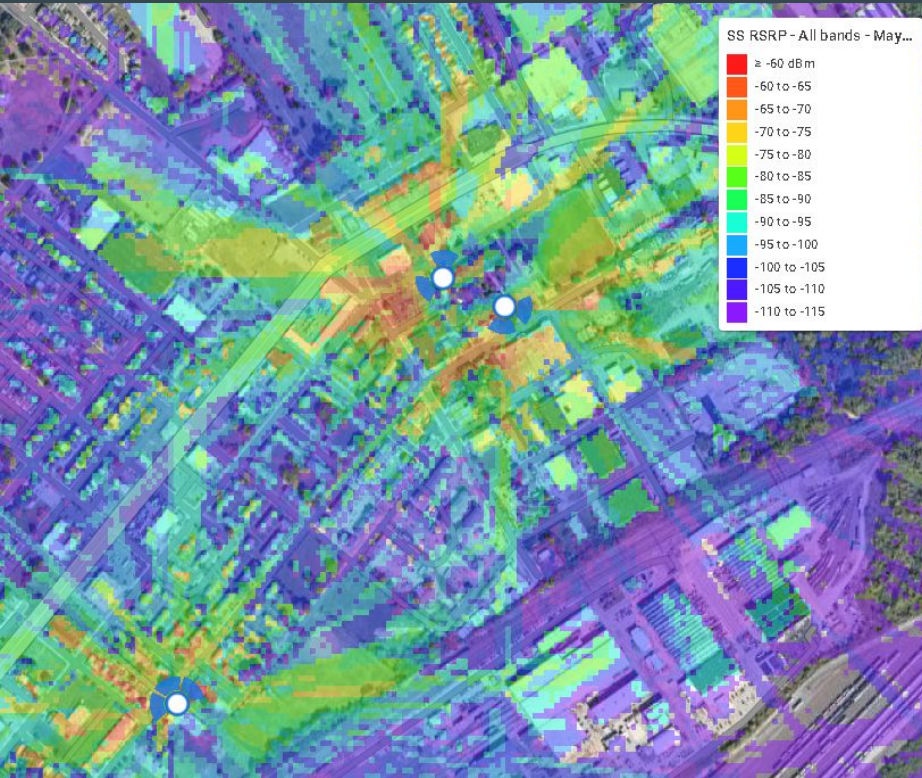
An engineer requests a report from a web form. Apps Script pulls the KPI data, builds the charts and returns the formatted output, with nothing installed locally.

Raspberry Pi and Testing Applications

Raspberry Pi as a portable test host: spectrum capture, throughput runs and data collection driven from a browser on the device itself, so no laptop is needed in the field.

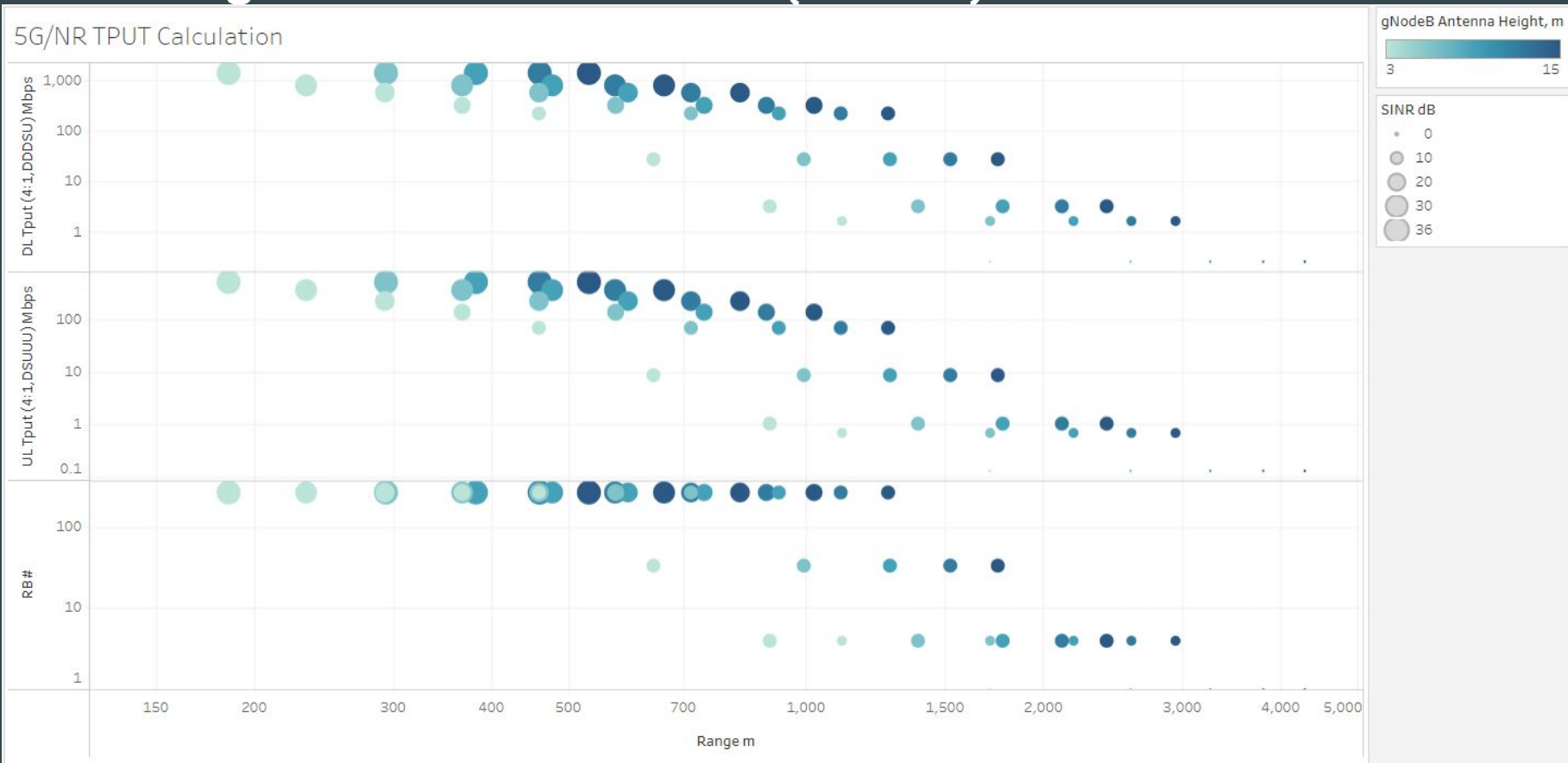


RF Design



Designing private Networks and DAS with a range of indoor and outdoor 5G/LTE solutions in the USA, UK, Mexico, etc.
Providing 4G/5G coverage plots, equipment lists, hardware installation locations.

Link Budget & TPUT Calculations (Tableau)



Example Link Budget Calculation based on 3GPP TR 38.901

Band	MIMO layers	DL BW	Mode	MCS index table selection 3GPP 38.214 (depends on DCI, C-RNTI, etc)	Modulation Type (QPSK-2, 16QAM-4, 64QAM-6,	modulation order	tpu for selected PRB count	UE assigned PRB	Target SINR	coverage range, m	Propagation model	LoS coverage	Reception sensitivity dBm	LoS signal level at outdoor receiver (Link Budget), dBm	gNodeB antenna height, m	Ue transmit power
n71	4	20	FDD	28	64QAM-6	6	60.993	19	25	1,270	Rural Macro 3D-RMA LoS/NLoS	TRUE	-78.66	-78.61	30	23
n71	4	20	FDD	16	16QAM-4	4	60.903	41	12	1,780	Rural Macro 3D-RMA LoS/NLoS	TRUE	-88.32	-88.19	30	23
n71	4	20	FDD	8	QPSK-2	2	60.476	89	-1	2,475	Rural Macro 3D-RMA LoS/NLoS	TRUE	-97.95	-97.82	30	23
n71	4	20	FDD	8	QPSK-2	2	6.795	10	-1	3,360	Rural Macro 3D-RMA LoS/NLoS	TRUE	-107.45	-107.08	30	23

LOS/NLOS	Pathloss [dB], f_c is in GHz and d is in meters, see note 6	Shadow fading std [dB]	Applicability range, antenna height default values
LOS	$PL_{\text{RMA-LOS}} = \begin{cases} PL_1 & 10\text{m} \leq d_{2D} \leq d_{BP} \\ PL_2 & d_{BP} \leq d_{2D} \leq 10\text{km} \end{cases}, \text{ see note 5}$ $PL_1 = 20 \log_{10}(40\pi d_{3D} f_c / 3) + \min(0.03h^{1.72}, 10) \log_{10}(d_{3D}) - \min(0.044h^{1.72}, 14.77) + 0.002 \log_{10}(h) d_{3D}$ $PL_2 = PL_1(d_{BP}) + 40 \log_{10}(d_{3D} / d_{BP})$	$\sigma_{\text{SF}} = 4$ $\sigma_{\text{SF}} = 6$	$h_{\text{BS}} = 35\text{m}$ $h_{\text{UT}} = 1.5\text{m}$ $W = 20\text{m}$ $h = 5\text{m}$ h = avg. building height W = avg. street width

Example TPUT Calculation based on 3GPP TS 38.306

SINR	MIMO layers	DL BW	Mode	Direction	MCS index table selection 3GPP 38.214 (depends on DCI, C-RNTI, etc)	Modulation Type (QPSK-2, 16QAM-4, 64QAM-8, 256QAM-8, 1024QAM-10)	modulation order	Rmax, Value depends on the type of coding from 3GPP 38.214 (Target code Rate R / 1024)	OH(j) -overhead for control channels (3GPP 38.306)	$\mu(0)=15\text{kHz}$	number of PRB (3GPP 38.104)	$T_{ps}(j) = (10^{-3})/(14 \cdot 2^{\mu})$ – average OFDM symbol duration in a subframe for $\mu(i)$ value for normal cyclic prefix	scaling factor	Supported max data rate Mbps
~+25	4	20	FDD	DL	28	64QAM-6	6	0.92578125	0.14	0	19	0.00007142857143	1	60.99
~+12	4	20	FDD	DL	16	16QAM-4	4	0.642578125	0.14	0	41	0.00007142857143	1	60.90
~-1	4	20	FDD	DL	8	QPSK-2	2	0.587890625	0.14	0	89	0.00007142857143	1	60.48
~-1	4	20	FDD	DL	8	QPSK-2	2	0.587890625	0.14	0	10	0.00007142857143	1	6.80

****5G Theoretical Examples**** (n71/SCS 15kHz/20MHz/4T2R/Rural Macro 3D-RMA ****LoS**** -- SINR values vary by vendor -- see notes below)

Lower Utilization



Higher Utilization

19 RBs deliver 61 Mbps at SINR ~+25dB: MCS-28 (64QAM), range 1,270m
 41 RBs deliver 61 Mbps at SINR ~+12dB: MCS-16 (16QAM), range 1,780m
 89 RBs deliver 61 Mbps at SINR ~ -1dB: MCS-8 (QPSK), range 2,475m

Better Quality



Worse Quality or Longer range

For NR, the approximate data rate for a given number of aggregated carriers in a band or band combination is computed as follows.

$$\text{data rate (in Mbps)} = 10^{-6} \cdot \sum_{j=1}^J \left(v_{\text{Layers}}^{(j)} \cdot Q_m^{(j)} \cdot f^{(j)} \cdot R_{\text{max}} \cdot \frac{N_{\text{PRB}}^{BW(j), \mu} \cdot 12}{T_s^{\mu}} \cdot (1 - OH^{(j)}) \right)$$

Comparing multi Local LLM for 3GPP Calculations – [link-1](#) [link-2](#)

To calculate the maximum data rate based on 3GPP TS 38.306, we use the standard formula for peak throughput.

1. Parameters and Constants

Based on your configuration, we have the following values:

```
* **Number of MIMO Layers ($v$):** $4$
* **Modulation Order ($Q_m$):** For 64 QAM, $Q_m = 6$ bits per symbol.
* **Coding Rate ($R$):** The coding rate is determined by the MCS (Modulation and Coding Scheme). For **MCS 28** with 64 QAM, the 3GPP TS 38.214 table typically assigns a coding rate of approximately **0.925**.
* **Resource Blocks ($N_{PRB}$):** $19$
* **Subcarriers per RB:** $12$
* **Numerology ($\mu = 0$):** $15 \text{ kHz}$ Subcarrier Spacing (SCS).
* In $15 \text{ kHz}$ numerology, $1 \text{ slot} = 14 \text{ symbols}$ and the slot duration is $1 \text{ ms}$ (meaning there are $1000 \text{ slots per second}$).
* **Overhead ($\text{OH}$):** $0.14$ (meaning we multiply by $1 - 0.14 = 0.86$).
```

Step D: Apply Overhead

Finally, we subtract the overhead to find the net data rate.

```
$$\text{Net Data Rate} = \text{Gross Rate} \times (1 - \text{OH})$$
$$\text{Net Data Rate} = 70,862,400 \times (1 - 0.14)$$
$$\text{Net Data Rate} = 70,862,400 \times 0.86 = 60,941,664 \text{ bps}$$
```

3. Final Result

Converting the result to Megabits per second (Mbps):

```
$$\text{Max Data Rate} \approx \mathbf{60.94 \text{ Mbps}}$$
```

Summary Table:

Parameter	Value
Total REs/s	$3,192 \times 10^6$
Bits per RE (effective)	22.25
Overhead Factor	0.86
Total Throughput	60.94 Mbps

Gemma4:31b -> 60.94 Mbps

>>> End a message (/? for help)

One question with a checkable answer: 4x4 MIMO, MCS 28, 64QAM, 0.14 overhead, 15 kHz numerology, 19 RBs, supported max data rate under 3GPP TS 38.306.

Three local models, several runs each. Only Gemma 4 31B returned 60.94 Mbps, matching my own implementation of the formula.

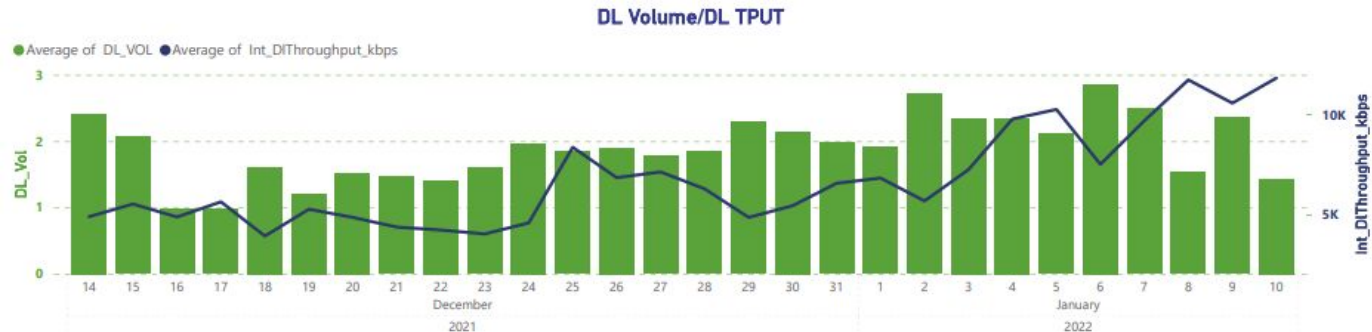
The arithmetic was not what failed — the other two could not recall Rmax from TS 38.214 Table 5.1.3.1-1 and used a code rate that does not exist for that MCS index, one of them while citing the table correctly by name.

Run locally at temperature 0 where the runtime allowed, across different runtimes and quantizations — a first check rather than a controlled comparison.

KPI Report

Microsoft Power BI

KPI collection over Amos:
uv sls_username > \$username1
pmxnhe .
^Acc_ErabAttempt\$|^Acc_InitialERabEstabSuccRate\$|^Acc_InitialERabEstabFailCount\$|^Acc_InitialERabEstabSuccRateQci1\$|^Acc_InitialERabEstabFailCountQci1\$|^Ret_ERabRetainabilityRate\$|^Ret_ERabRetainabilityCount\$|^DL_VOL\$|^Int_DIThroughput_kbps\$|^Tot_Downtime\$ -f
/home/shared/\$username1/FORMULA_v01.txt -tz -7 -o csv -m 48



Parameter Audit

M0	parameter	KGET value	GC Value	comparison
ENodeBFunction=1,EUtranCellFDD=INL02	threshXHighQ	18	['12', '14']	FALSE
ENodeBFunction=1,EUtranCellFDD=INL02	a2CriticalThresholdRsrp	-126	-126	TRUE
ENodeBFunction=1,EUtranCellFDD=INL02	inhibitA2SearchConfig	2	2	TRUE
ENodeBFunction=1,EUtranCellFDD=INL02	ulTrigActive	TRUE	FALSE	FALSE
ENodeBFunction=1,EUtranCellFDD=INL02	connectedModeMobilityPrio	5	1	FALSE
ENodeBFunction=1,EUtranCellFDD=INL02	threshXHighQ	18	14	FALSE
ENodeBFunction=1,EUtranCellFDD=INL02	connectedModeMobilityPrio	5	1	FALSE
ENodeBFunction=1,EUtranCellFDD=INL02	threshXHighQ	18	14	FALSE
ENodeBFunction=1,EUtranCellFDD=INL02	threshXLowQ	14	(18, 22)	FALSE
ENodeBFunction=1,EUtranCellFDD=INL02	tReselectionEutra	1	2	FALSE
ENodeBFunction=1,EUtranCellFDD=INL02	connectedModeMobilityPrio	5	1	FALSE
ENodeBFunction=1,EUtranCellFDD=INL02	threshXHighQ	18	14	FALSE
ENodeBFunction=1,EUtranCellFDD=INL02	tReselectionEutra	1	2	FALSE
ENodeBFunction=1,EUtranCellFDD=INL02	threshXHighQ	18	14	FALSE
ENodeBFunction=1,EUtranCellFDD=INL02	connectedModeMobilityPrio	5	1	FALSE
ENodeBFunction=1,EUtranCellFDD=INL02	threshXHighQ	18	['12', '14']	FALSE
ENodeBFunction=1,EUtranCellFDD=INL02	a2CriticalThresholdRsrp	-126	-126	TRUE
ENodeBFunction=1,EUtranCellFDD=INL02	inhibitA2SearchConfig	2	2	TRUE
ENodeBFunction=1,EUtranCellFDD=INL02	ulTrigActive	TRUE	FALSE	FALSE
ENodeBFunction=1,EUtranCellFDD=INL02	connectedModeMobilityPrio	5	1	FALSE
ENodeBFunction=1,EUtranCellFDD=INL02	threshXHighQ	18	14	FALSE
ENodeBFunction=1,EUtranCellFDD=INL02	connectedModeMobilityPrio	5	1	FALSE
ENodeBFunction=1,EUtranCellFDD=INL02	threshXHighQ	18	14	FALSE
ENodeBFunction=1,EUtranCellFDD=INL02	threshXLowQ	14	(18, 22)	FALSE
ENodeBFunction=1,EUtranCellFDD=INL02	tReselectionEutra	1	2	FALSE
ENodeBFunction=1,EUtranCellFDD=INL02	connectedModeMobilityPrio	5	1	FALSE
ENodeBFunction=1,EUtranCellFDD=INL02	threshXHighQ	18	14	FALSE
ENodeBFunction=1,EUtranCellFDD=INL02	tReselectionEutra	1	2	FALSE
ENodeBFunction=1,EUtranCellFDD=INL02	threshXHighQ	18	14	FALSE

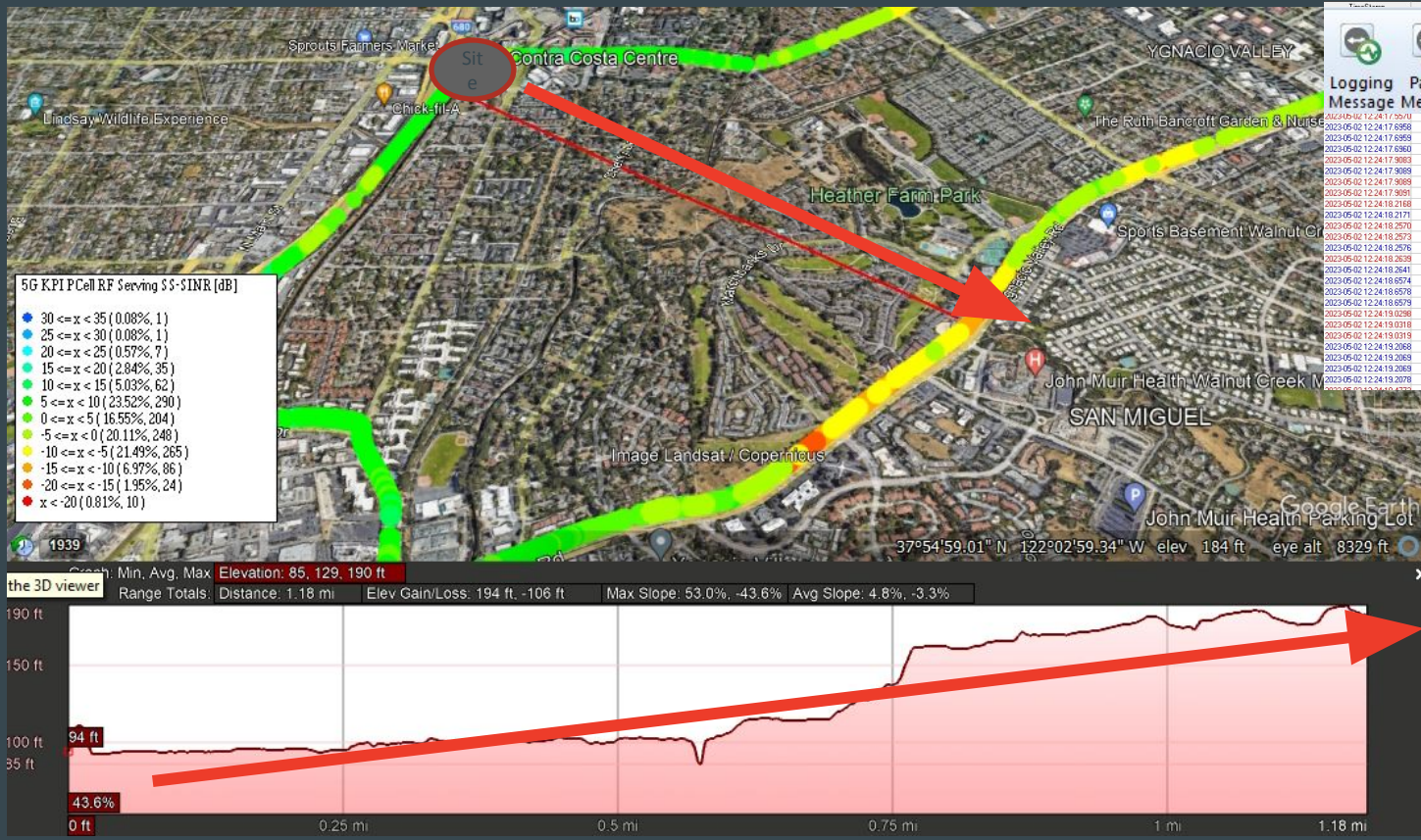
KGET output diffed against the golden configuration. Red rows are drift from the template; each is corrected on the network, or the baseline is updated to match intent.

Cluster Testing



Network status, connectivity, static performance and mobility testing across the cluster. Confirms every cell is transmitting as designed, and that coverage, SINR, throughput, MIMO usage, drop rate and handover success match the design targets before the cluster is accepted.

Network Optimization & Acceptance Tests



Time/Process	IPs	IP	APP/APP	NME/APP	Interface	Message
					3GPP NAS	Registration complete
					5G NR RRC	ulInformationTransfer
					5G NR RRC	dlInformationTransfer
					3GPP NAS	Configuration update request
					3GPP NAS	PDU session establishment request
					3GPP NAS	UL NAS transport
					5G NR RRC	ulInformationTransfer
					5G NR RRC	maintainBearerInfo
					5G NR RRC	systemInformationBlockType1
					3GPP NAS	PDU session establishment request
					3GPP NAS	UL NAS transport
					5G NR RRC	ulInformationTransfer
					5G NR RRC	dlInformationTransfer
					3GPP NAS	reRegistrationComplete
					3GPP NAS	reConfiguration
					5G NR RRC	reConfigurationComplete
					3GPP NAS	PDU session establishment accept
					5G NR RRC	reConfigurationComplete
					3GPP NAS	reConfigurationComplete
					3GPP NAS	UL NAS transport
					3GPP NAS	PDU session establishment accept
					5G NR RRC	maintainBearerInfo
					5G NR RRC	dlInformationTransfer
					3GPP NAS	reConfigurationComplete
					3GPP NAS	reConfiguration
					5G NR RRC	reConfigurationComplete
					SIP	REGISTER
					SIP	REGISTER
					SIP	REGISTER
					SIP	401 Unauthorized
					SIP	401 Unauthorized
					SIP	401 Unauthorized
					SIP	REGISTER
					SIP	REGISTER
					SIP	REGISTER
					SIP	REGISTER
					SIP	REGISTER

Indoor and outdoor drive testing across several toolchains, then post-processing: terrain profile read against serving SINR, signaling flow beside the log, and the optimization actions that follow from both.

Example Ericsson Kget file parser via Python

GitHub [link](#)

```
for file0 in onlyfiles:
    text_file = open(os.path.join(mydirectoryoutput,file0+"parsed.txt"), "w")
    with open(os.path.join(mydirectoryinput,file0), 'r') as file01:
        text_file.write("row#MOclass#MOclassUpdate#MO#parameter#value\n")
        for num, line in enumerate(file01, 1):
            #for line in file01:
                if(line[:1] is "=" or line[:5] is "Proxy"):
                    #print line
                    continue
                MO = re.match("MO\s+(SubNetwork=.*)\s+(\w+)\s+(.*)", line)
                if MO:
                    MO01 = MO.group(1) + " " + MO.group(2) + "=" + MO.group(3)
                    MOclass01 = MO.group(2)
                    MOclassUp = MO.group(2) + "=" + MO.group(3)
                    continue

#####
    struct01 = re.match("Struct\s+(\w+)\s+has\s+(\d+)\s+members\s+:$", line) #Str
    if struct01:
        struct01flag= int(struct01.group(2))
        parameterPart_01_1 = struct01.group(1)
        continue
    #struct01A = re.match("/".join(["\s+>>>\s+\d+\s+(\w+)\s+=\s+(\d+)", "\s+>>>
    struct01A = re.match("\s+>>>\s+\d+\s+(\w+)\s+=\s+(.*)$", line)
    # >>> 1.mcc = 310
```

Python Parser

```
MO
=====
ExternalEUTranCellFDDId      310410-550320-10
activePlmnList[2]
>>> Struct[0] has 3 members:
>>> 1.mcc = 310
>>> 2.mnc = 410
>>> 3.mncLength = 3
activeServiceAreaId[0]
additionalFreqBandList[1]    25
commonSFAllocList[0]
createdBy                     0 (ANR)
dlChannelBandwidth           20000
eutranFrequencyRef           SubNetwork=ONRM_ROOT_M
freqBand                     2
isRemoveAllowed              true
```

input

	MO	parameter	value	
BFunction=:	ExternalEUTranCellFDD=	ExternalEUTranCellFDDId		10
BFunction=:	ExternalEUTranCellFDD=	activePlmnList[0].mcc		10
BFunction=:	ExternalEUTranCellFDD=	activePlmnList[0].mnc		10
BFunction=:	ExternalEUTranCellFDD=	activePlmnList[0].mncLength		3
BFunction=:	ExternalEUTranCellFDD=	activePlmnList[1].mcc		13
BFunction=:	ExternalEUTranCellFDD=	activePlmnList[1].mnc		00
BFunction=:	ExternalEUTranCellFDD=	activePlmnList[1].mncLength		3
BFunction=:	ExternalEUTranCellFDD=	createdBy		0
BFunction=:	ExternalEUTranCellFDD=	dlChannelBandwidth		20000
BFunction=:	ExternalEUTranCellFDD=	eutranFrequencyRef	SubNetwork	
BFunction=:	ExternalEUTranCellFDD=	freqBand		2
BFunction=:	ExternalEUTranCellFDD=	isRemoveAllowed	TRUE	
BFunction=:	ExternalEUTranCellFDD=	lastModification		1
BFunction=:	ExternalEUTranCellFDD=	lbEUTranCellOffloadCapacity		1000
BFunction=:	ExternalEUTranCellFDD=	lbmld		1
BFunction=:	ExternalEUTranCellFDD=	localCellId		10
BFunction=:	ExternalEUTranCellFDD=	noOfTxAntennas		2
BFunction=:	ExternalEUTranCellFDD=	physicalLayerCellId		407
BFunction=:	ExternalEUTranCellFDD=	physicalLayerCellIdGroup		135
BFunction=:	ExternalEUTranCellFDD=	physicalLayerSubCellId		2
BFunction=:	ExternalEUTranCellFDD=	remoteCellState		1
BFunction=:	ExternalEUTranCellFDD=	remoteCellStateEranUIComp		1
BFunction=:	ExternalEUTranCellFDD=	remoteCellStateInterEnbUIComp		0
BFunction=:	ExternalEUTranCellFDD=	removingMonitoringStart		2018
BFunction=:	ExternalEUTranCellFDD=	tac		34645
BFunction=:	ExternalEUTranCellFDD=	timeOfCreation		2018
BFunction=:	ExternalEUTranCellFDD=	timeOfLastModification		2018
BFunction=:	ExternalEUTranCellFDD=	ulChannelBandwidth		20000
BFunction=:	ExternalEUTranCellFDD=	uniqueCellId		140881930
BFunction=:	ExternalEUTranCellFDD=	zzzTemporaryExt1		-2000000000

output

Parses Ericsson KGET exports into a flat MO, parameter and value table, which is what makes bulk diffing and the parameter audit possible.

Scripting on Ericsson **Amos** language – [link](#)

```

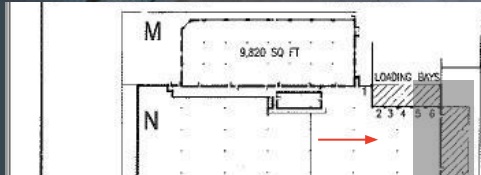
XXX004X3> $sec1 = `moshelledir/moshell $!paddress 'lt ^SectorCarrier$,y;s+;hgetc sectorcarrier reserve' nolog | awk '{if ($1 ~ /Sector/) {printf "%s#", $1 $
$sec1 = SectorCarrier#SectorCarrier=1;[2]=EUTranCellFDD=XXX004X3_7A_X U1CompGroup=2#SectorCarrier=2;[2]=EUTranCellFDD=XXX004X3_7B_X U1CompGroup=2#SectorCar
XXX004X3> pmxtd SectorCarrier pmRadioRecInterferencePwrBrPrb | awk -v wait1="$pm wait" -v sec="$sec1" -v count="0" -v sample="0" -v branch="" -v key1="0

```

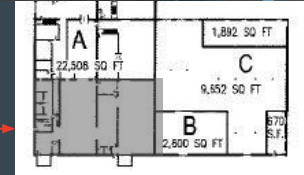
Object	Date/Time:	2018-07-03 -- 03:03	Info	Value	info
SectorCarrier=1,PmUInterferenceReport=1			RSSI	-118.287	;[2]=EUTranCellFDD=XXX004X3_7A_X U1CompGroup=2
SectorCarrier=1,PmUInterferenceReport=2			RSSI	-117.603	;[2]=EUTranCellFDD=XXX004X3_7A_X U1CompGroup=
SectorCarrier=2,PmUInterferenceReport=1			RSSI	-117.466	;[2]=EUTranCellFDD=XXX004X3_7B_X U1CompGroup=2
SectorCarrier=2,PmUInterferenceReport=2			RSSI	-118.418	;[2]=EUTranCellFDD=XXX004X3_7B_X U1CompGroup=2
SectorCarrier=3,PmUInterferenceReport=1			RSSI	-117.858	;[2]=EUTranCellFDD=XXX004X3_7C_X U1CompGroup=2
SectorCarrier=3,PmUInterferenceReport=2			RSSI	-117.989	;[2]=EUTranCellFDD=XXX004X3_7C_X U1CompGroup=2
SectorCarrier=4,PmUInterferenceReport=1			RSSI	-117.726	;[2]=EUTranCellFDD=XXX004X3_9A_X U1CompGroup=1
SectorCarrier=4,PmUInterferenceReport=2			RSSI	-118.542	;[2]=EUTranCellFDD=XXX004X3_9A_X U1CompGroup=1
SectorCarrier=4,PmUInterferenceReport=3			RSSI	-117.541	;[2]=EUTranCellFDD=XXX004X3_9A_X U1CompGroup=1
SectorCarrier=4,PmUInterferenceReport=4			RSSI	-117.951	;[2]=EUTranCellFDD=XXX004X3_9A_X U1CompGroup=1
SectorCarrier=5,PmUInterferenceReport=1			RSSI	-119.177	;[2]=EUTranCellFDD=XXX004X3_9B_X U1CompGroup=1
SectorCarrier=5,PmUInterferenceReport=2			RSSI	-119.12	;[2]=EUTranCellFDD=XXX004X3_9B_X U1CompGroup=1
SectorCarrier=5,PmUInterferenceReport=3			RSSI	-118.589	;[2]=EUTranCellFDD=XXX004X3_9B_X U1CompGroup=1
SectorCarrier=5,PmUInterferenceReport=4			RSSI	-118.659	;[2]=EUTranCellFDD=XXX004X3_9B_X U1CompGroup=1
SectorCarrier=6,PmUInterferenceReport=1			RSSI	-118.655	;[2]=EUTranCellFDD=XXX004X3_9C_X U1CompGroup=1
SectorCarrier=6,PmUInterferenceReport=2			RSSI	-118.779	;[2]=EUTranCellFDD=XXX004X3_9C_X U1CompGroup=1
SectorCarrier=6,PmUInterferenceReport=3			RSSI	-117.97	;[2]=EUTranCellFDD=XXX004X3_9C_X U1CompGroup=1
SectorCarrier=6,PmUInterferenceReport=4			RSSI	-117.723	;[2]=EUTranCellFDD=XXX004X3_9C_X U1CompGroup=1

Site Walk

→ Photo line of view



Site inspection before design. Directional photography tied to a line of view on the floor plan, with fiber access, power access and obstructions recorded, so the design is built against what is on the ground.



Field CX Inspection/COP



Thank you for taking the time to read it.

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